

Hemanth Kumar Galam

Backend Engineer | Distributed Systems

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PROFESSIONAL SUMMARY

Software Engineer with 5+ years building high-concurrency, event-driven distributed systems and real-time data pipelines, with growing experience integrating agentic AI into industrial applications.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python

Backend & APIs: Node.js, Express.js, FastAPI, REST APIs, WebSockets, OpenAPI/Swagger, RabbitMQ

Authentication & Security: JWT, RBAC, MFA, OAuth2, Rate Limiting, OWASP Practices

Databases & Caching: PostgreSQL, MongoDB, Redis

Cloud & DevOps: AWS (Lambda, S3, RDS), Docker, Kubernetes, CI/CD, Jenkins, Linux, Git

Observability & Monitoring: Splunk, Prometheus, Grafana

PROFESSIONAL EXPERIENCE

Senior Backend Engineer

SS&C Intralinks (via Ensar Solutions)

May 2023 -- Present

Remote / Germany

- Developed core backend features for VIA V2, the modern successor to a 15-year-old legacy platform, including transactional REST API middleware, RBAC, MFA, and notification and alerting services.
- Migrated over **300 TB** of files from NFS to **AWS** using an asynchronous pipeline powered by **Argo Workflows** across multiple data centers.
- Architected an event-driven design with asynchronous processing using **RabbitMQ** to efficiently distribute system load under high concurrency.
- Implemented Inbox and Outbox event-driven patterns to guarantee transactional consistency and idempotency across distributed microservices.
- Led API design and documentation initiatives, facilitating seamless integration and onboarding for global enterprise customers such as Barclays, AXA, and Bank of America.

Software Engineer (Research Assistant)

Fraunhofer IPA

Jan 2024 -- Mar 2026

Stuttgart, Germany

- Developed a **Full-stack Node.js** application for real-time visualization of robotic inference data, bridging the gap between **ROS2** and web-based monitoring.
- Architected a WebSocket-driven pipeline to stream high-frequency skeletal data, achieving low-latency live inference display in an Angular frontend.
- Designed the backend database architecture and data schemas to optimize the storage and retrieval of real-time robotic telemetry for long-term analysis.

Software Engineer Intern

Rohde & Schwarz GmbH

Apr 2023 -- Sep 2023

Munich, Germany

- Developed a custom RL-encoding compression package in **JavaScript**, reducing map-loading latency by **40%**.
- Engineered UI features for the MAP application utilizing Node.js and Angular.
- Followed Agile/Scrum methodologies with a focus on Clean Code and maintained **95% test coverage** using Jest.

Senior Software Engineer

Ensar Solutions Pvt. Ltd.

Dec 2020 -- Apr 2023

Hyderabad, India

- Designed and developed the system architecture for access management and CRM workflows serving **200,000+ daily active users**.
- Practiced Agile/Scrum for iterative delivery and managed 40+ REST APIs optimized for high-concurrency workloads.
- Optimized **PostgreSQL** queries and indexing strategies, reducing API response times by **40%**.

PROJECTS

Migrator ETL Tool

- Architected a scalable ETL tool for high-integrity, seamless data migration between SQL (**PostgreSQL**) and NoSQL (**MongoDB**) systems.

TO-DO GPT

- Developed an AI-powered task management application utilizing GPT models to automate task categorization and prioritization based on natural language input.

Multi-LLM FastAPI

- Built a high-performance **FastAPI** backend that integrates multiple Large Language Models (LLMs) via a single unified interface for optimized prompt routing.

EDUCATION

Master of Science in Electrical and Computer Engineering

Oct 2022 -- Aug 2026

University of Stuttgart, Germany

- **Master Thesis (Grade: 1.3):** "Domain-Specific Action Recognition for HRC using Synthetic Data."
- Developed a real-time pipeline on edge devices using GCN and Transformers for industrial collaboration.
- **Research Thesis (Grade: 1.0):** "ML-based Predictive Modeling for Thermal Runaway."
- Integrated Regression, Classification, and YOLO-based detection for battery safety monitoring.
- **Focus:** Machine Learning and Deep Learning (GPA: 2.1 / Good).

Bachelor of Technology in Electrical and Electronics Engineering

Jul 2017 -- Mar 2021

Jawaharlal Nehru Technological University, India

ACHIEVEMENTS

Winner -- sustainATHON 2023 (Stuttgart, Germany) •

3rd Place -- Bayerwald Hackathon (Deggendorf, Germany)

LANGUAGES

German (A2), English (C1 - Fluent)